

Лабораторная №7

Обоснование выбора индексов (запрос №1)

Индекс	Причина
idx_ord_fac_pk ON ordered_facility (order_id, driver_id)	GROUP BY orf.order_id, orf.driver_id
idx_order_time ON "order" (cancelled, order_time)	WHERE NOT o.cancelled AND MAKE_TIME(EXTRACT(HOUR FROM o.order_time)::INTEGER, 0, 0) BETWEEN ...
idx_payment3 ON payment (order_id, driver_id) WHERE type_id = 3	JOIN payment pa ON pa.order_id = o_ext.order_id AND pa.driver_id = o_ext.driver_id AND pa.type_id = 3
idx_order_driver_end_time_nocanc ON "order" (driver_id, cancelled, end_time);	WHERE oo.driver_id = o.driver_id AND NOT oo.cancelled AND oo.end_time < o.end_time
idx_review_order_user ON review (order_id, user_id);	JOIN review r ON r.order_id = oo.order_id AND r.user_id <> oo.driver_id

Сравнение показателей (запрос №1)

Показатель	До	После
total cost	8891.75..1521785.91	8891.75..768493.42
execution time	70812.610 ms	14720.136 ms
buffers	shared hit=68 dirtied=1	shared hit=115 read=1

Изменения в плане выполнения (запрос №1)

-> Bitmap Heap Scan on "order" o (cost=615.88..3319.24 rows=1 width=40) (actual time=1.805..13.916 rows=498.36 loops=56) Recheck Cond: (NOT cancelled) Filter: ((EXTRACT(dow FROM order_time) = os.week_day) AND (make_time((EXTRACT(hour FROM order_time))::integer, 0, '0')::double precision <= os.opening_time)) Rows Removed by Filter: 27410 Heap Blocks: exact=85512 Buffers: shared hit=91452 read=108
-> Bitmap Index Scan on idx_order_time (cost=0.00..615.88 rows=27679 width=0) (actual time=1.617..1.617 rows=27908.00 loops=56) Index Cond: (cancelled = false) Index Searches: 56 Buffers: shared hit=5940 read=108
-> Index Only Scan using idx_payment3 on payment pa (cost=0.29..8.31 rows=1 width=16) (actual time=0.002..0.002 rows=0.85 loops=27908) Index Cond: ((order_id = o.order_id) AND (driver_id = o.driver_id)) Heap Fetches: 0 Index Searches: 27908 Buffers: shared hit=55817
-> Bitmap Heap Scan on "order" oo (cost=4.51..67.38 rows=17 width=16) (actual time=0.010..0.028 rows=28.78 loops=23836) Recheck Cond: ((driver_id = o.driver_id) AND (NOT cancelled) AND (end_time < o.end_time)) Heap Blocks: exact=675909 Buffers: shared hit=726820
-> Bitmap Index Scan on idx_order_driver_end_time_nocanc (cost=0.00..4.50 rows=17 width=0) (actual time=0.006..0.006 rows=28.78 loops=23836) Index Cond: ((driver_id = o.driver_id) AND (cancelled = false) AND (end_time < o.end_time)) Index Searches: 23836 Buffers: shared hit=50911
-> Index Scan using idx_review_order_user on review r (cost=0.29..9.85 rows=2 width=20) (actual time=0.002..0.002 rows=0.67 loops=685972) Index Cond: (order_id = oo.order_id) Filter: (user_id <> oo.driver_id) Rows Removed by Filter: 1 Index Searches: 685972 Buffers: shared hit=1993584

Обоснование выбора индексов (запрос №3)

Индекс	Причина
idx_user_id_role ON "user" (user_id, role_id);	RIGHT JOIN "user" u ON o.passenger_id = u.user_id AND u.role_id = 1
idx_order_pass ON "order" (cancelled, end_time, passenger_id)	WHERE NOT o.cancelled AND o.end_time IS NOT NULL AND o.passenger_id = u.user_id
те же, что и для запроса №1	

Сравнение показателей (запрос №3)

Показатель	До	После
total cost	19475.46..103437680.77	19428.30..26002539.00
execution time	291042.923 ms	16943.846 ms
buffers	shared hit=83 dirtied=1	shared hit=236 read=10

Изменения в плане выполнения (запрос №3)

-> Index Only Scan using idx_user_id_role on "user" u_1 (cost=0.29..346.94 rows=11110 width=12) (actual time=0.015..0.941 rows=11110.00 loops=1)
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=3 read=42
-> Index Scan using idx_order_pass on "order" o_1 (cost=0.41..395.51 rows=1 width=48) (actual time=0.184..0.541 rows=2.72 loops=9242)
Index Cond: ((cancelled = false) AND (end_time IS NOT NULL) AND (passenger_id = u.user_id))
Index Searches: 9242
Buffers: shared hit=1189458 read=126
-> Index Only Scan using idx_payment3 on payment pa_1 (cost=0.29..8.31 rows=1 width=16) (actual time=0.002..0.003 rows=0.95 loops=25105)
Index Cond: ((order_id = o_1.order_id) AND (driver_id = o_1.driver_id))
Heap Fetches: 0
Index Searches: 25105
Buffers: shared hit=50211
-> Nested Loop (cost=0.70..277.86 rows=1 width=4) (actual time=0.186..0.290 rows=0.85 loops=23836)
Buffers: shared hit=1651253
-> Index Scan using idx_order_pass on "order" oo (cost=0.41..267.75 rows=1 width=16) (actual time=0.156..0.286 rows=1.26 loops=23836)
Index Cond: ((cancelled = false) AND (end_time < o_1.end_time) AND (passenger_id = o_1.passenger_id))
Index Searches: 23836
Buffers: shared hit=1564102
-> Index Scan using idx_review_order_user on review r (cost=0.29..10.09 rows=2 width=20) (actual time=0.002..0.003 rows=0.68 loops=29983)
Index Cond: (order_id = oo.order_id)
Filter: (user_id <> oo.passenger_id)
Rows Removed by Filter: 1
Index Searches: 29983
Buffers: shared hit=87151